



Electroporation (Transformation)

Designing Life with AI

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Description

Electroporation is one of the methods used for transformation, alongside chemical transformation (which involves calcium chloride and heat shock). In this method, a high-voltage electric pulse is applied to create temporary pores in the cell membrane, allowing plasmid DNA to enter.

Material

- Competent cells (*E. coli* or *P. putida*)
- Plasmid DNA
- LB or S.O.C medium
- Sterile saline (optional, for *P. putida* plating)
- Plate with antibiotic

Steps

For *E. coli* strains

1. Mix 2 μ L of plasmid preparation with 100 μ L of competent cells in a sterile electroporation cuvette.
2. Set the electroporator to 2500 V and apply a single pulse (insertion time should be 4–6 ms).
3. Immediately add 1 mL of LB or S.O.C medium to the cuvette.
4. Incubate the cells at 37 °C and 800 rpm for 40 to 60 minutes.
5. Plate the cells on agar plates containing the appropriate antibiotic using sterile glass beads.
6. Incubate the plates overnight at 37 °C in a static incubator to allow colony growth.

For *P. putida* strains

1. Transfer 50 μ L of electrocompetent cells within a prechilled electroporation cuvette, on ice.
2. Mix your plasmid (500–1000 ng total quantity for recombination i.e. Tn5, Tn7 or whole plasmid integration; 10–50 ng for replicating plasmid) with your electrocompetent cells. Leave on ice for 15 min.
3. Electroporation is done at 2 kV.
4. Immediately after electroporation, add 1 mL of SOC medium into the electroporation cuvette. No need to keep on ice anymore.
5. Transfer the cell suspension into a sterile eppendorf tube, and incubate at 35 °C. For Tn5 and Tn7 insertion, typical incubation time is 1–2 h, and up to 3 hours for whole plasmid integration into the genome.
6. Plate 100 μ L of your cell suspension onto one plate, centrifuge the 900 μ L remaining and resuspend within 100 μ L SOC, LB or saline. Plate the concentrated cell suspension onto another plate.
7. Incubate the plates for 16–24 h at 30 °C.